# Spotify Music Track Analysis

Aatash Pestonjamasp, Edward Lee, Liang Yang
Danni Ma, Yashas Anilkumar

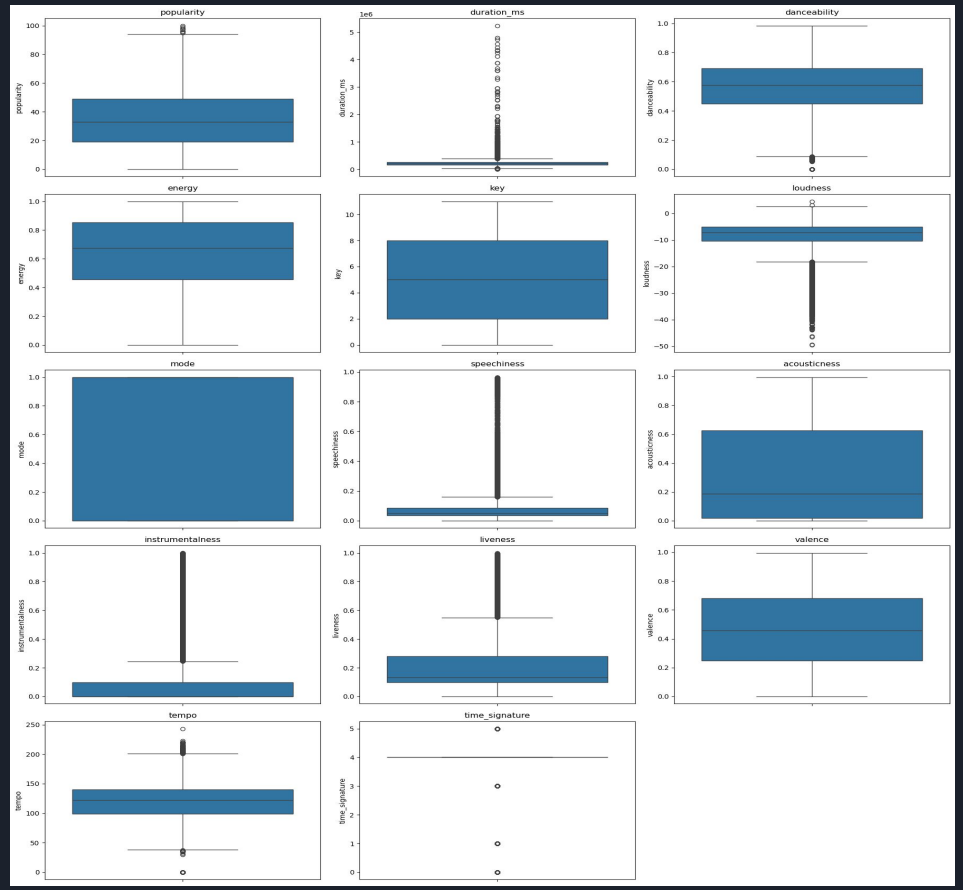
# Data Overview & Methodology - Spotify Tracks Dataset

```
RangeIndex: 114000 entries, 0 to 113999
Data columns (total 21 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   Unnamed: 0        114000 non-null  int64
 1   track_id          114000 non-null  str
 2   artists           113999 non-null  str
 3   album_name        113999 non-null  str
 4   track_name        113999 non-null  str
 5   popularity        114000 non-null  int64
 6   duration_ms       114000 non-null  int64
 7   explicit          114000 non-null  bool
 8   danceability      114000 non-null  float64
 9   energy            114000 non-null  float64
 10  key               114000 non-null  int64
 11  loudness          114000 non-null  float64
 12  mode              114000 non-null  int64
 13  speechiness       114000 non-null  float64
 14  acousticness      114000 non-null  float64
 15  instrumentalness  114000 non-null  float64
 16  liveness          114000 non-null  float64
 17  valence           114000 non-null  float64
 18  tempo             114000 non-null  float64
 19  time_signature    114000 non-null  int64
 20  track_genre       114000 non-null  str
dtypes: bool(1), float64(9), int64(6), str(5)
```

- Dataset of Spotify tracks from over 125 different genres in .csv
  - https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset/data
- Each track has audio feature associated, like
  - **Track_id**, **artists**, **album_name**, **track_name**, **loudness**, **key**, **track_genre**,
  - **Popularity**: Mostly based on total number of plays and how recent plays are
  - **Danceability**: Combination of tempo, rhythm, beat regularity, etc
  - **Tempo**: Overall estimated tempo in beats per minute (BPM); pace
  - **Speechiness**: Detects the presence of spoken words in a track.
  - ...
- Methodology:
  - **Coding & Data Analysis**: Jupyter Notebook, Pandas, Numpy,
  - **Data Visualization**: Seaborn & pyplot (matplotlib)

# 1. Do tracks have any abnormal values/outliers?

- The dataset contains **114000** rows in total
- Only **1** row with null value
- No **fully duplicate** rows
- Did remove songs being listed in multiple genres
    - Worried about correlation matrix having bias from multi-contribution

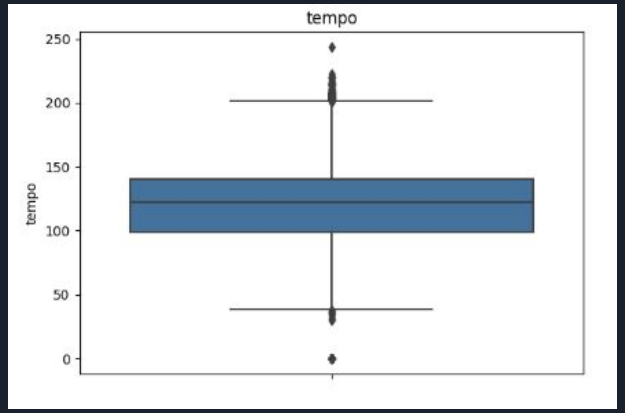


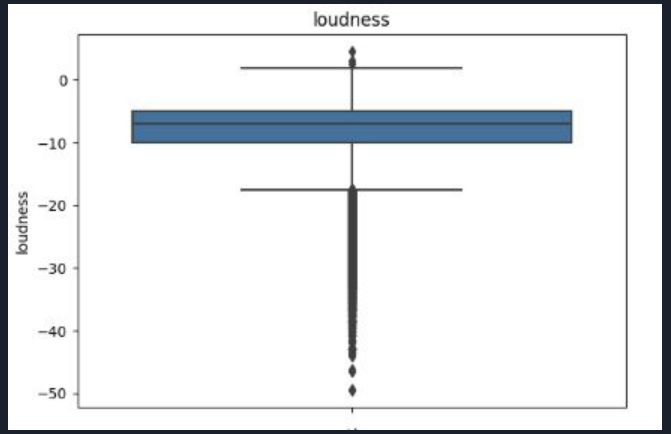
# 1. Do tracks have any abnormal values/outliers?

- We see outliers for each numerical feature and their percentage

| Anomaly Type | Count | % of Data | Interpretation / Impact |
|---|---|---|---|
| Instrumentalness outliers | 19613 | 21.90% | Songs that are 100% instrumental (unusual for mainstream music) |
| Speechiness outliers | 10644 | 11.90% | Heavily speech-focused (podcasts, rap, spoken word) |
| Time signature outliers | 10197 | 11.40% | Non-standard time signatures (e.g., 5/4, 7/8) |
| Liveness outliers | 6981 | 7.80% | Live recordings and concert audio |
| Loudness outliers | 5026 | 5.60% | Extremely quiet recordings |
| Duration outliers | 4225 | 4.70% | Very short interludes or very long mixes |
| Tempo ≈ 0 | 157 | 0.20% | ABNORMAL: No detectable beat (ambient, white noise) |
| Extreme popularity | 11 | 0.01% | RARE: Megahits with >95 popularity score |
| Loudness < -45 dB | 4 | <0.01% | VERY ABNORMAL: Nearly silent recordings |

# 1. Do tracks have any abnormal values/outliers?



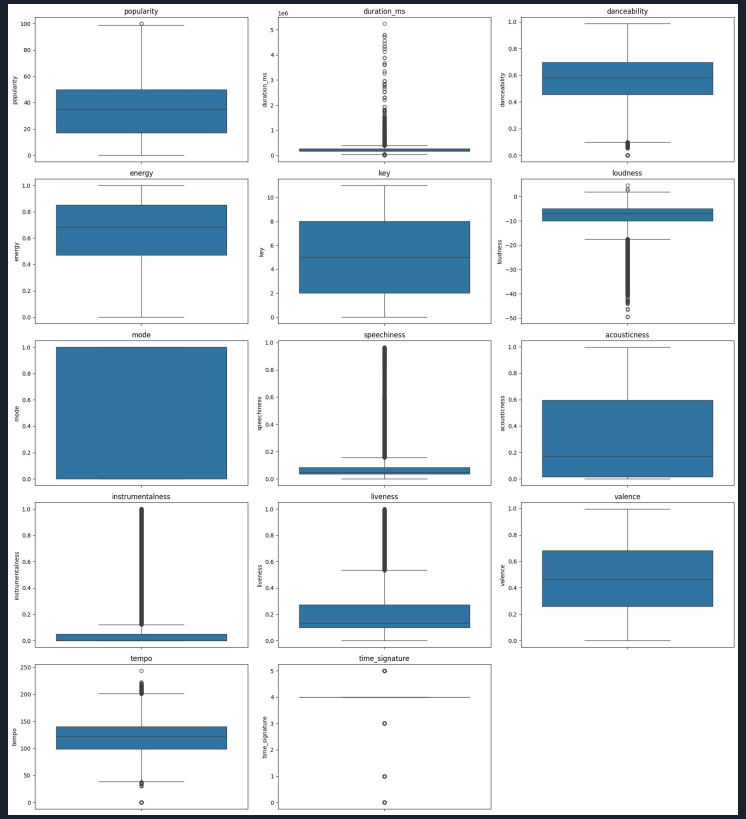

**Most Possible Abnormal data:**

- **<u>Tempo</u>** = **0** BPM: totally **157** rows
  - Failed to detect a beat
    - Ambient/drone music
    - Spoken-word tracks
    - Silence-heavy recordings

- **<u>Loudness</u>** < **-45** dB: totally **4** rows
  - Measured in dBFS (decibels relative to full scale),
    - 0 dB is the max digital amplitude
    - Values are negative

  - <u>Loudness < -45dB:</u> Extremely quiet,
    - Silence
    - Recording artifacts
    - Metadata errors

# 2. Do songs tend to have specific characteristics?



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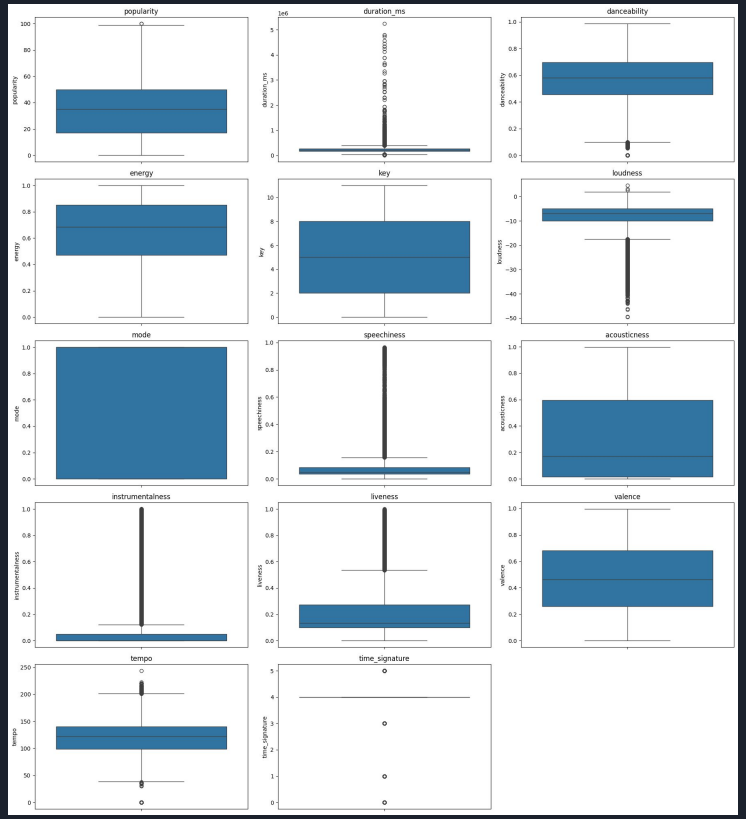


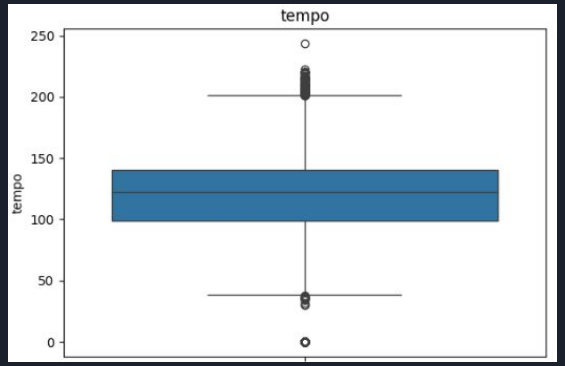

Short answer:

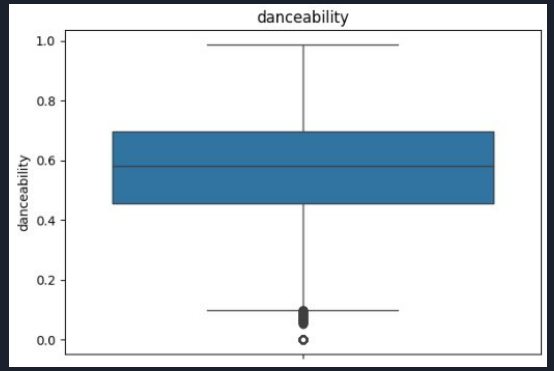
Yes!

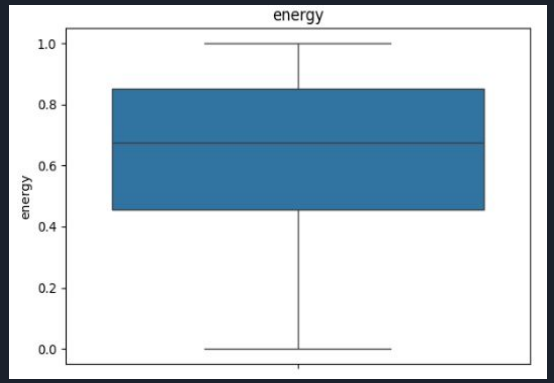
# 2. Do songs tend to have specific characteristics?

**Specific Examples:**

- **Tempo:** Tends to be roughly within 100-150 bpm
- **Danceability:** Tend to be more danceable than not
- **Valence (positivity):** Wide range
- **Energy:** Tends to be higher



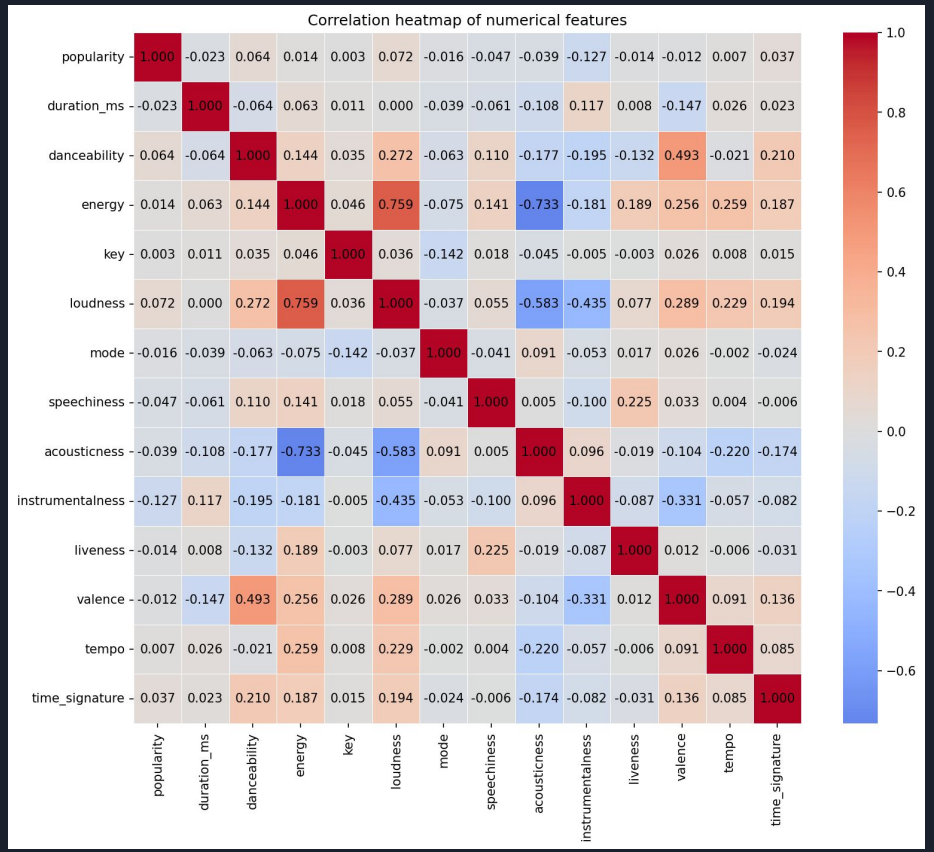





# 3. Are there any features which are notably correlated with each other?

## Specific Examples:

- Loudness and Energy
- Acousticness and Energy
- Acousticness and Loudness
- Valence and Danceability
- Etc.



# 4a. What features affect (or don't affect) popularity?

| Feature | Coefficient | P-Value | R-Squared |
|---|---|---|---|
| instrumentalness | -10.787567 | 0.000000e+00 | 0.037994 |
| speechiness | -13.204728 | 8.158655e-129 | 0.007235 |
| danceability | 7.652255 | 2.630571e-98 | 0.005500 |
| loudness | 0.257252 | 1.183672e-95 | 0.005349 |
| explicit | 4.249673 | 6.891067e-78 | 0.004338 |
| liveness | -4.590729 | 3.681112e-46 | 0.002531 |
| energy | -2.436029 | 3.248559e-22 | 0.001169 |
| tempo | -0.011636 | 4.435381e-08 | 0.000373 |
| mode | -0.307011 | 2.076694e-02 | 0.000067 |
| acousticness | -0.369690 | 5.311217e-02 | 0.000047 |
| key | 0.009647 | 5.915774e-01 | 0.000004 |
| valence | -0.128383 | 5.988858e-01 | 0.000003 |

- **Question/ Method:**
  - Is there is a simple linear relationship between popularity and features of a song?
  - Ordinary Least Squares
- Regress popularity on independent features to see which feature has the best predictive quality
- P-values for each regression are small enough to be statistically significant and not just be random chance
- Overall results show that instrumentalness alone is the best predictor for popularity, explaining ~3.8% of the variation in popularity

# 4b. Do groups of features explain the data better than individual ones?

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             popularity   R-squared:                       0.070
Model:                            OLS   Adj. R-squared:                  0.070
Method:                 Least Squares   F-statistic:                     867.1
Date:                Tue, 03 Mar 2026   Prob (F-statistic):               0.00
Time:                        19:55:22   Log-Likelihood:            -3.4363e+05
No. Observations:               80293   AIC:                         6.873e+05
Df Residuals:                   80285   BIC:                         6.874e+05
Df Model:                           7
Covariance Type:            nonrobust
=====================================================================================
                        coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------
Intercept            41.3642      0.276    149.915      0.000      40.823      41.905
explicit[T.True]      5.0868      0.236     21.574      0.000       4.625       5.549
instrumentalness    -12.5968      0.202    -62.234      0.000     -12.994     -12.200
speechiness         -20.2145      0.582    -34.749      0.000     -21.355     -19.074
danceability          9.3548      0.412     22.695      0.000       8.547      10.163
liveness             -2.1139      0.331     -6.387      0.000      -2.763      -1.465
energy               -3.2773      0.257    -12.751      0.000      -3.781      -2.774
valence              -7.1485      0.286    -24.991      0.000      -7.709      -6.588
==============================================================================
Omnibus:                     1515.452   Durbin-Watson:                   0.669
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              960.020
...
==============================================================================
```

**Forward model selection:**

- Add features with highest adjusted R-squared
- Only keep if improves the adjusted R-squared of new model
- Some features that didn't increase the adjusted r-squared value were loudness and tempo

**Some guesses to why:**

- Loudness correlates with energy so they predict in the same way
- Tempo is highly dependent on the genre (a confounding variable), and some genres may just be unpopular in general

# 4b. Do groups of features explain the data better than individual ones? (cont.)

Another approach:

- Consider all combinations of features
- Saw best r-squared  and model complexity with 11 features
- Likely works out since this dataset is fairly simple (considering 12 features max)

| Num_Features | Features | R-Squared | Adj_R-Squared | AIC | BIC |
|---|---|---|---|---|---|
| 11 | (explicit, danceability, energy, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo) | 0.071684 | 0.071557 | 687163.855068 | 687275.376321 |
| 12 | (explicit, danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo) | 0.071686 | 0.071547 | 687165.672574 | 687286.487265 |
| 10 | (explicit, danceability, energy, loudness, mode, speechiness, instrumentalness, liveness, valence, tempo) | 0.071569 | 0.071453 | 687171.815241 | 687274.043056 |
| 11 | (explicit, danceability, energy, key, loudness, mode, speechiness, instrumentalness, liveness, valence, tempo) | 0.071571 | 0.071444 | 687173.602641 | 687285.123894 |
| 10 | (explicit, danceability, energy, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence) | 0.071535 | 0.071420 | 687174.724170 | 687276.951985 |
| 11 | (explicit, danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence) | 0.071538 | 0.071410 | 687176.526872 | 687288.048125 |

# 5. Looking into one specific genre (kpop)

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             popularity   R-squared:                       0.298
Model:                            OLS   Adj. R-squared:                  0.294
Method:                 Least Squares   F-statistic:                     81.03
Date:                Tue, 10 Mar 2026   Prob (F-statistic):           6.09e-71
Time:                        23:37:25   Log-Likelihood:                -3633.8
No. Observations:                 960   AIC:                             7280.
Df Residuals:                     954   BIC:                             7309.
Df Model:                           5
Covariance Type:            nonrobust
====================================================================================
                       coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------
Intercept           90.2830      2.868     31.478      0.000      84.654      95.912
explicit[T.True]    -9.6339      2.521     -3.821      0.000     -14.582      -4.686
loudness             2.0730      0.168     12.322      0.000       1.743       2.403
acousticness       -14.2349      1.634     -8.711      0.000     -17.442     -11.028
energy             -20.9271      2.979     -7.026      0.000     -26.773     -15.082
speechiness         11.3089      4.343      2.604      0.009       2.785      19.833
==============================================================================
Omnibus:                      307.225   Durbin-Watson:                   1.288
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1795.547
Skew:                          -1.336   Prob(JB):                         0.00
Kurtosis:                       9.144   Cond. No.                         94.2
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

- Trying to understand the variance within just one category
- Adjusted R-squared of 0.294
- K-pop market may select for loud songs which are non explicit, and have more vocals (speechiness)

# Thank you!

1. *Abnormalities:* Tracks may vary but aren't necessarily invalid as tracks.
2. *Features values:* Some attributes take on typical values
3. *Correlation:* Some features (like loudness and energy) can strongly correlate
4. *Popularity:* Individual features can explain popularity, but not as well
   - Groups of features can improve on explaining popularity
5. *Genres:* Focusing on a specific genre (like kpop) can yield better insights

# Thank you!